AERO50001 Aerodynamics 2 Problem Set #2

Extra practice: Normal shocks, convergent-divergent nozzles

2.1 Isentropic nozzle

★☆☆ 15-18 mins

Helium, at $T_0 = 400$ K, enters a nozzle isentropically. At section 1, where $A_1 = 0.1\ \mathrm{m}^2$, a pitot static tube arrangement measures a stagnation pressure of $150\ \mathrm{kPa}$ and a static pressure of $123\ \mathrm{kPa}$.
For Helium, $\gamma = 1.66$, $R = 2077\ \mathrm{J/kg \cdot K}$.

(a) Estimate M_1, the Mach number at section 1.

(b) Estimate T_1, the temperature at section 1.

(c) Estimate the mass flow rate.

(d) Estimate A^*, the area at the throat.

(a) At section 1, for isentrophic flow:

$$\frac{p_{01}}{p_1} = \left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\frac{\gamma}{\gamma-1}}$$

$$\therefore M_1 = 0.4988$$

(b) $T_0 = T_{01} = 400K$

$$\frac{T_{01}}{T_1} = 1 + \frac{\gamma-1}{2} M_1^2 \qquad \therefore T_1 = 369.65K$$

(c) $$\dot{m} = \rho_1 u_1 A_1 = \rho_1 M_1 a_1 A_1 = \frac{p_1}{RT_1} M_1 \sqrt{\gamma R T_1}\, A_1$$

$$= \frac{p_1 M_1 A_1 \sqrt{\gamma}}{\sqrt{RT_1}} = 9.021\ \mathrm{kg/s}$$

(d) $$\left(\frac{A_1}{A^*}\right)^2 = \frac{1}{M_1^2}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2} M_1^2\right)\right]^{\frac{\gamma+1}{\gamma-1}}$$

$$\therefore A^* = 0.0756\ \mathrm{m}^2$$

2.2 Converging-diverging nozzle

★★☆ 15-20 mins

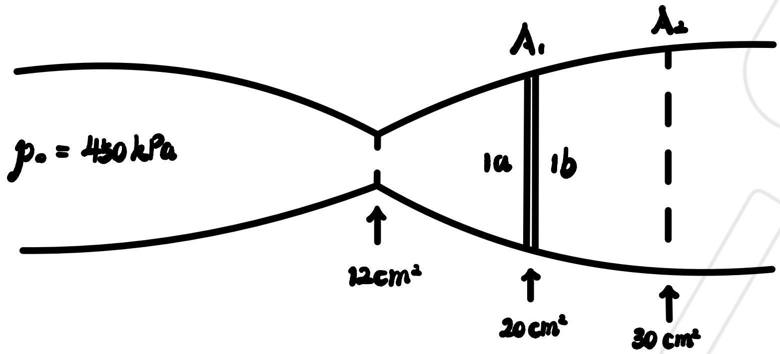
Air, supplied by a reservoir at $450\ \mathrm{kPa}$ flows through a converging-diverging nozzle whose throat area is $12\ \mathrm{cm}^2$. A normal shock stands at section 1, where $A_1 = 20\ \mathrm{cm}^2$. Still further downstream is section 3, where $A_3 = 30\ \mathrm{cm}^2$.

(a) Compute the pressure just downstream of the shock at section 1.

(b) Estimate A_3^*, the area of the second throat required to choke the flow at section 3.

(c) Estimate the Mach number at section 3, M_3.

(d) Estimate the pressure P_3 at section 3.



(a) Just upstream of the normal shock, for isentropic process:

$$\frac{A_1}{A^*} = \frac{5}{3} \qquad \therefore M_{1a} = 1.98 \text{ according to the figure}$$

$$\therefore \frac{p_0}{p_{1a}} = 7.5849 \qquad \therefore p_{1a} = \frac{p_0}{7.5849}$$

$$\therefore \frac{p_{1b}}{p_{1a}} = 4.407 \qquad \therefore p_{1b} = \frac{4.407}{7.5849} p_0 = 261.46\mathrm{kPa}$$

(b) $M_{1b} = 0.5808$ according to the figure

$$\therefore \left(\frac{A_1}{A_3^*}\right)^2 = \frac{1}{M_{1b}^2}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M_{1b}^2\right)\right]^{\frac{\gamma+1}{\gamma-1}} = 1.4687$$

$$\therefore A_3^* = 16.5\mathrm{cm}^2$$

(c) $\frac{A_3}{A_3^*} = 1.818$

$\therefore M_3 = 0.34$ according to the figure

(d) $\frac{p_{01b}}{p_0} = 0.7302 \qquad \therefore p_{01b} = 0.7302 p_0$

$\frac{p_{01b}}{p_3} = 1.0833 \qquad \therefore p_3 = \frac{0.7302}{1.0833} p_0 = 303.32\,kPa$

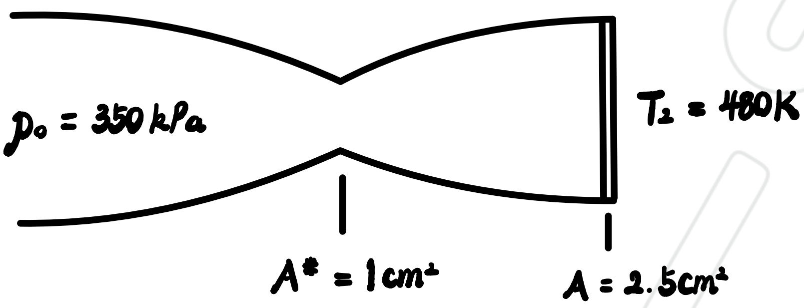
2.3 Nozzle with shock

★☆☆ 10-15 mins

A tank at $350\ \mathrm{kPa}$ feeds air ($\gamma = 1.4$) into a nozzle with a throat area of $1\ \mathrm{cm}^2$ and an exit area of $2.5\ \mathrm{cm}^2$. A normal shock occurs at the exit of the nozzle, and just behind the normal shock the temperature is $480\ \mathrm{K}$.

(a) Calculate the temperature in the tank.

(b) Calculate the pressure behind the shock.



(a) $\frac{A}{A^*} = 2.5$

$\therefore M_1 = 2.45$, $\frac{p_2}{p_1} = 6.836$ and $\frac{\rho_2}{\rho_1} = 3.273$

$\because \frac{T_2}{T_1} = (\frac{p_2}{p_1})(\frac{\rho_1}{\rho_2}) = \frac{6.836}{3.273}$ $\quad \therefore T_1 = 229.82K$

$\because \frac{T_0}{T_1} = 2.2005$ $\quad \therefore T_0 = 505.72K$

(b) $\frac{p_0}{p_1} = 15.806$ $\quad \therefore p_1 = 22.143kPa$

$\frac{p_2}{p_1} = 6.836$ $\quad \therefore p_2 = 151.37kPa$